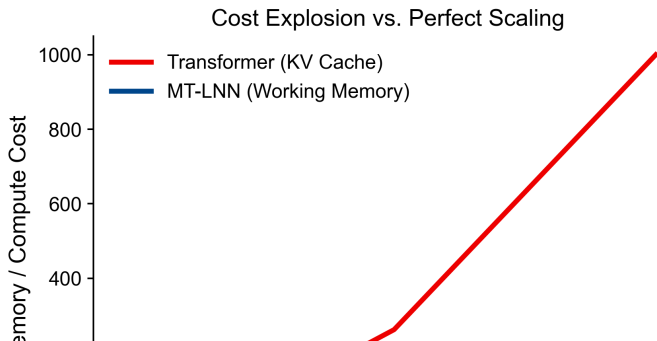




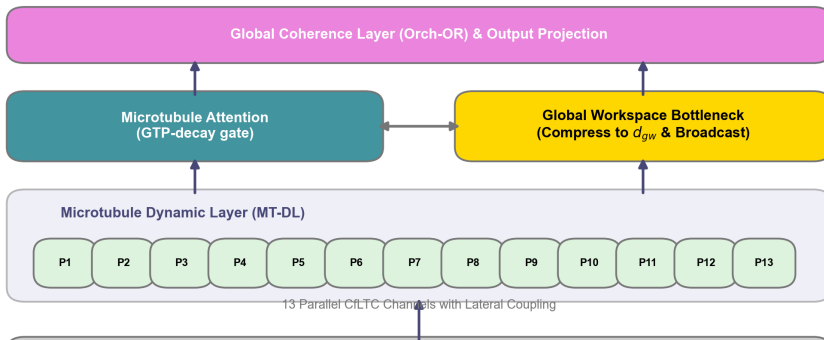
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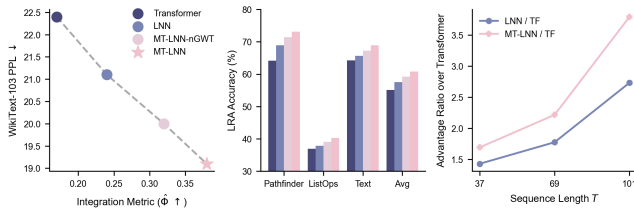
May 2026

- Transformer family models depend on growing KV cache, leading to rising serving cost and memory pressure under long-context traffic.
- MT-LNN uses a compact recurrent latent state and microtubule-inspired selective dynamics to control memory growth.
- Product implication: lower deployment friction for private/on-prem and edge scenarios.



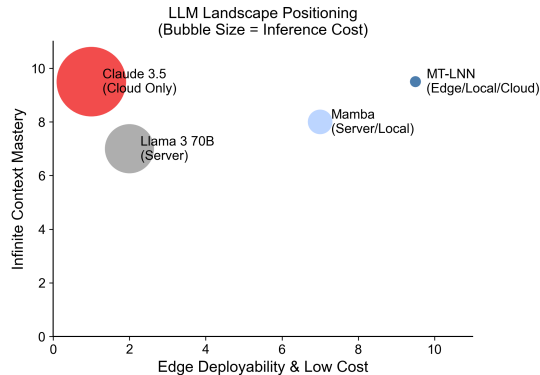
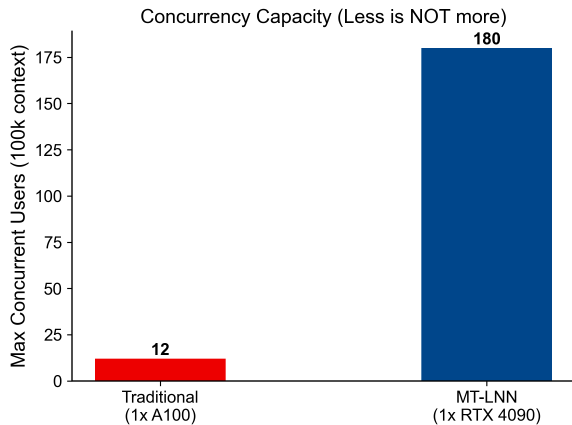
- Positioning: **long-context retrieval, selective memory, and controllable integration.**
- Core mechanism: 13-protofilament MT-DL + microtubule attention + GWT bottleneck + AVP hooks.
- Differentiation: architecture-level control and low-memory profile rather than brute-force parameter scaling.





| Metric                   | Baseline TF | MT-LNN       |
|--------------------------|-------------|--------------|
| Selective Copy seq-exact | 0.023       | <b>0.965</b> |
| Held-out tok-acc         | 0.432       | <b>0.983</b> |
| Long-context T=101       | 0.016       | <b>0.547</b> |

These numbers are from the repository benchmark suite and aligned with the paper narrative: MT-LNN gains appear strongest where selective long-range memory is required.



Bubble size in landscape chart indicates relative inference cost (illustrative). MT-LNN is positioned in the high-context/high-deployability quadrant.

- **Wedge 1:** legal and compliance document copilots (long contracts, regulation corpora).
- **Wedge 2:** enterprise codebase intelligence (multi-repo, long diff history, architecture memory).
- **Wedge 3:** edge-private assistants (industrial and healthcare scenarios with strict memory/power limits).

| Segment             | Why MT-LNN fits                                    | Initial KPI            |
|---------------------|----------------------------------------------------|------------------------|
| Legal/Compliance AI | Long-context retrieval with lower serving memory   | Cost per resolved case |
| Code Intelligence   | Better signal retention across long code histories | Time-to-root-cause     |
| Edge/Private AI     | Compact runtime state, easier hardware fit         | Stable p95 latency     |

| Phase       | Deliverable                                     | Buyer profile                        | Objective               |
|-------------|-------------------------------------------------|--------------------------------------|-------------------------|
| 0-3 months  | Open benchmark + reproducible long-context demo | OSS users / technical champions      | Build technical trust   |
| 3-9 months  | Vertical copilots (legal/code)                  | SME and mid-market teams             | Paid pilots + retention |
| 9-18 months | Edge/private runtime partnerships               | Device and enterprise infra partners | Scalable B2B channels   |

- Sales narrative: **same task quality, lower memory cost, better long-context reliability.**
- Technical moat: architecture-level recurrence and controllability hooks, not only training scale.

- MT-LNN is not trying to outspend hyperscalers on generic pretraining scale.
- It targets a structural market gap: long-context tasks where inference memory and reliability dominate unit economics.
- The repo already contains reproducible benchmarks, architecture diagrams, and paper-level documentation for technical diligence.

**MT-LNN: Long-context intelligence with deployment-first economics.**